

**MARK SCHEME for the October/November 2009 question paper
for the guidance of teachers**

9709 MATHEMATICS

9709/71

Paper 71, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

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Mark Scheme Notes

Marks are of the following three types:

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.
- Note: B2 or A2 means that the candidate can earn 2 or 0.
B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.
- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only – often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR –1	A penalty of MR –1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures – this is regarded as an error in accuracy. An MR–2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA –1	This is deducted from A or B marks in the case of premature approximation. The PA –1 penalty is usually discussed at the meeting.

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1 $X \sim B(180, 0.02) \sim \text{Po}(3.6)$ $P(X < 4) = e^{-3.6} (1 + 3.6 + 3.6^2 / 2 + 3.6^3 / 6)$ $= 0.515$	B1 M1 A1 [3]	Poisson with mean 180×0.02 Poisson attempt with their λ allow end errors Correct answer SR ₁ Use of Bin scores B1 only for ans 0.514 SR ₂ Use of Normal scores B1 only for 0.479
2 $1.96 \times \frac{1.5}{\sqrt{n}} < \frac{1}{2}$ $n = 35$	B1 B1 M1 A1 [4]	$1.96 \times \frac{1.5}{\sqrt{n}}$ seen Confidence interval halved Solving an equation in their z , 1.5, n (2 and sq rt not needed) Correct answer (condone $n \geq 35$)
3 (i) $P(\bar{W} > 51) = P\left(z > \frac{51 - 48.5}{12.4 / \sqrt{5}}\right)$ $= 1 - \Phi(0.451)$ $= 1 - 0.674$ $= 0.326$	M1 M1 A1 [3]	Standardising with 51 and mean 48.5 Standardising using $\sqrt{5}$ Correct answer
(ii) $z = 1.5$ or 1.499 $\frac{51.6 - 48.5}{(12.4 / \sqrt{n})} = 1.5$ $\sqrt{n} = 6$ $n = 36$	B1 M1 M1 A1 [4]	1.5 or 1.499 seen Standardising must have $\sqrt{n}$ (no cc) Attempt to solve equation with $\sqrt{n}$, their z in correct answer
4 (i) $P(X > 4) = 1 - P(0, 1, 2, 3, 4)$ $= 1 - e^{-1.8} \left(1 + 1.8 + \frac{1.8^2}{2} + \frac{1.8^3}{3!} + \frac{1.8^4}{4!} \right)$ $= 1 - 0.9635$ $= 0.036(4)$ This is < 0.05 and so $X > 4$ is in the critical region $P(4) = e^{-1.8} \left(\frac{1.8^4}{4!} \right) = 0.0723$	M1 M1 A1 A1ft B1 [5]	Adding at least 3 relevant Poisson terms Poisson expression for $P(X > 4)$ (oe implied by later working) Correct prob 0.036 (or 0.96 subject to later working) Correct comparison and statement identifying CR (ft their prob < 0.05) Verification that $X = 4$ is not in the cr region
(ii) $P(\text{Type II error}) = P(X = 0, 1, 2, 3, 4)$ $= e^{-2.3} \left(1 + 2.3 + \frac{2.3^2}{2} + \frac{2.3^3}{3!} + \frac{2.3^4}{4!} \right)$ $= 0.916$	B1 M1 A1 [3]	Correct region Poisson expression $P(0, 1, 2, 3, 4)$ Correct answer

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5 (i) $\int_0^{\pi/4} k \cos x \, dx = 1$ $[k \sin x]_0^{\pi/4} = 1$ $k \sin(\pi/4) = 1 \Rightarrow k/\sqrt{2} = 1$ $k = \sqrt{2}$ AG	M1 A1 [2]	Equating to 1 and attempt to integrate with limits Correct answer legit obtained (no decimals seen)
(ii) $\int_{0.4}^{\pi/4} k \cos x \, dx = [k \sin x]_{0.4}^{\pi/4}$ $= 1 - k \sin(0.4)$ $= 0.449$	M1 A1 [2]	Attempt to integrate from 0.4 to $\pi/4$ o.e. Correct answer
(iii) $\int_0^{Q3} k \cos x \, dx = 0.75$ $[k \sin x]_0^{Q3} = 0.75$ $k \sin Q3 - 0 = 0.75$ $Q3 = 0.559$	M1 M1 A1 [3]	Equation with integral on one side and 0.75 on the other o.e. Attempt to solve their integral for Q3 Correct answer
(iv) ${}^5C_3 \times (0.25)^3 \times (0.75)^2$ $= 0.0879$ (45/512)	M1 A1 [2]	Binomial expression involving 5C_3, 0.25 and 0.75 Correct answer
6 (i) $\bar{x} = 14.8$ (890/60 oe) $s^2 = \frac{1}{59} \left(13780 - \frac{890^2}{60} \right)$ $= 9.80$	B1 M1 A1 [3]	Correct answer Substituting in formula from book, o.e. Correct answer
(ii) $H_0: \mu = 15.2$ $H_1: \mu < 15.2$ $P(\text{Type I error}) = 0.1$ (10%) Say the photographer has fewer discards when she doesn't	B1 B1 B1ft [3]	Correct H_1 and H_0 Correct answer o.e. must be related to question. No contradictions. ft their H_1
(iii) Test statistic $z = \frac{14.83 - 15.2}{\sqrt{\frac{9.802}{60}}}$ $= -0.915$ $CV \, z = \pm 1.282$ Not enough evidence to support photographer's claim.	M1 A1 M1 A1ft [4]	Standardising must have $\sqrt{60}$ Correct z (± 0.91 to 0.92) or correct area 0.18 Valid comparison with correct CV must be + with + or – with – and consistent with their H_1 oe comparison of areas Correct conclusion ft their z and their CV No contradictions

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7	(i) $3C \sim N(990, 5.2^2 \times 3)$ ($= N(990, 81.12)$) $2W \sim N(1000, 7.1^2 \times 2)$ ($= N(1000, 100.82)$) $3C - 2W \sim N(-10, 181.94)$ $P((3C - 2W) < 0) = \Phi\left(\frac{0 - (-10)}{\sqrt{181.94}}\right)$ $= \Phi(0.741)$ $= 0.771$	B1	Correct mean for both 3 cans cola and 2 bottles water
		B1	Correct variance for both 3 cans cola and 2 bottles water
		M1	Correct method for mean and variance for $3C - 2W$ or vice versa
		M1	Standardising and using tables, need the sq root and area > 0.5
		A1	[5] Correct answer
	(ii) new drink $\sim N(910, 2 \times 5.2^2 + 0.5^2 \times 7.1^2)$ $\sim N(910, 66.68)$ $P(ND > 900) = 1 - P\left(z < \frac{900 - 910}{\sqrt{66.68}}\right)$ $= 1 - P(z < -1.225)$ $= \Phi(1.225)$ $= 0.8897$ (0.890)	B1	Correct mean for new drink
		B1	Correct variance for new drink
		M1	Standardising with sq rt and using tables
		A1	[4] Correct answer